

ADITI MAHAROR

AI/ML Research Intern Candidate | B.Tech Computer Science & Engineering, 2028

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OBJECTIVE

Computer Science undergraduate with hands-on experience building and deploying machine learning and LLM-based systems, including predictive modeling, prompt engineering, and data pipelines using Python and SQL. Seeking to contribute to Sony Research India's AI Research Internship on user behavior modeling and causal inference for OTT platforms by applying strong ML fundamentals, rapid prototyping ability, and experience reading and implementing research-driven approaches.

SKILLS

Languages: Python, C++, C, Go

AI / ML: Machine Learning, Deep Learning (TensorFlow/Keras), Large Language Models (LLMs), Prompt Engineering, AI Agents, Vector Databases, Model Evaluation

Data & Analysis: SQL, MySQL, SQLite, Large-Scale Dataset Processing, Data Preprocessing, Statistical Analysis

Frameworks / Backend: FastAPI, Django, React

Tools & Deployment: Git, GitHub, Docker, Render, Hugging Face, VS Code

Core CS: Data Structures & Algorithms, OOP, Problem Solving, Research Paper Implementation

PROJECTS

Casting Quality Defect Detection System (EdgeInspect) — TCOE IMC 2026 | TensorFlow/Keras, MobileNetV2, FastAPI

- Built and deployed a predictive deep learning model for real-time image-based quality classification, achieving 98.45% training and 99.16% test accuracy on a large-scale image dataset.
- Owned the end-to-end ML pipeline: image preprocessing (224x224), model loading and inference, and a FastAPI /predict endpoint; resolved a Keras version-mismatch bug via a weights-only model rebuild.
- Deployed the trained model via Docker on Render and hosted model weights on Hugging Face, validating the pipeline in a production-like environment.

AI Interview Agent — ABTalks Vibe Code Hackathon (2026) | Python, FastAPI, Claude LLM API

- Designed an adaptive interview system that models candidate confidence and competence using cohort-level behavioral signals to dynamically weight question selection, generating a quantitative 'calibration gap' metric.
- Implemented the LLM-integration and evaluation logic end-to-end in Python/FastAPI, iterating through structured experimentation and prompt refinement to improve response quality.

Deterministic Notification Routing Pipeline — HackerRank Orchestrate (AI Agent Challenge) | Python

- Developed a six-module pipeline modeling user interaction signals to predict the correct routing action per message, achieving 86.7% action accuracy on labeled evaluation data.
- Ranked 828 of 1,773 participants; iteratively tuned rule logic against ground-truth labels to improve prediction accuracy.

AI Placement Mentor (Ongoing) | Python, FastAPI, React, LLMs, Vector Database

- Architecting a personalization system matching resumes to 100+ companies and generating individualized learning roadmaps via semantic search over a vector database; building an LLM-based mock interview module with multi-parameter response evaluation.

Smart India Hackathon (SIH) 2025 — Disaster Preparedness & Response Education System

- Contributed to a gamified platform modeling user decision-making under emergency scenarios; presented to evaluators and received recognition for clear technical communication.

EDUCATION

B.Tech, Computer Science & Engineering — Birla Institute of Technology, Mesra (Off-Campus Patna) | 2024 – 2028 | CGPA: 8.52/10

Higher Secondary (Class 12) — St. Michael's High School, Patna | 2021 – 2023 | 93.4%

Matriculation (Class 10) — St. Michael's High School, Patna | 2011 – 2021 | 98% (Top 5% nationally, CBSE)

ACHIEVEMENTS

- NPTEL Elite — Top 1% (100% Score)
- Ranked 828 of 1,773 participants — HackerRank Orchestrate (AI Agent Challenge)
- 98% in CBSE Class 10 — Top 5% nationally
- Recognized for effective presentation of disaster-preparedness project at SIH 2025

CERTIFICATIONS

- Building Basic AI Agents and Custom GPTs — Udemy
- Python & Django Framework — Udemy
- AI Essentials: Introduction to AI
- HackerRank SQL (Advanced)

LEADERSHIP & EXTRA-CURRICULAR

- Volunteer & Co-organizer, Xordium 2025 — Coordinated logistics and participant support for technical and academic hackathon events.
- Member, Ignite Club (BIT Patna) — Contributed to content planning and digital engagement to improve club outreach.
- Volunteer, Treasury Club — Technika '24 — Supported event coordination and operational planning for college technical activities.